

Chapter 1 Assignment: Solution Report

Student Name: Mugabo Edson Student ID: 27502

Task 1: Student Profile System

This implementation handles unstructured terminal inputs via descriptive labels, dynamic scaling systems, mapping profiles, and automated timeline calculation formulas.

Source Code Implementation

```
full_name = input("Enter full name: ")
student_id = input("Enter student ID: ")
age = int(input("Enter age: "))
country = input("Enter country of origin: ")
current_year = int(input("Enter current year of study (1-4): "))
gpa = float(input("Enter current GPA: "))
major = input("Enter major/course of study: ")
remaining_years = 4 - current_year
graduation_year = 2026 + remaining_years
print(f"\n" + "="*40)
print(f"STUDENT PROFILE: {full_name.upper()}")
print(f"="*40)
print(f"{'Name':<25}: {full_name}")
print(f"{'Student ID':<25}: {student_id}")
print(f"{'Age':<25}: {age} years")
print(f"{'Country of Origin':<25}: {country}")
print(f"{'Current Year of Study':<25}: Year {current_year}")
print(f"{'Current GPA':<25}: {gpa:.2f}")
print(f"{'Major/Course':<25}: {major}")
print(f"{'Expected Graduation':<25}: {graduation_year}")
print(f"="*40)
```

Execution Terminal Output

```
=====
STUDENT PROFILE: MUGABO EDSON
=====
Name : Mugabo Edson
Student ID : 27502
Age : 21 years
Country of Origin : Rwanda
Current Year of Study : Year 2
Current GPA : 3.85
Major/Course : Computer Science
Expected Graduation : 2028
=====
```

Task 2: Advanced Calculator

Implements evaluation structures converting string parameters into primitive floats to route values across multiple arithmetic operators simultaneously.

Source Code Implementation

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
```

```

addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2
division = num1 / num2 if num2 != 0 else 0.0
floor_division = num1 // num2 if num2 != 0 else 0.0
modulus = num1 % num2 if num2 != 0 else 0.0
power = num1 ** num2
num1_parity = "Even" if num1 % 2 == 0 else "Odd"
num2_parity = "Even" if num2 % 2 == 0 else "Odd"
percentage = (num1 / num2) * 100 if num2 != 0 else 0.0
print(f"\n" + "="*40)
print(f"ARITHMETIC CALCULATIONS")
print(f"="*40)
print(f"Addition : {addition:.2f}")
print(f"Subtraction : {subtraction:.2f}")
print(f"Multiplication : {multiplication:.2f}")
print(f"Division : {division:.4f}")
print(f"Floor Division : {floor_division:.2f}")
print(f"Modulus : {modulus:.2f}")
print(f"Power Elevation : {power:.2f}")
print(f"="*40)
print(f"First Number Parity : {num1_parity}")
print(f"Second Number Parity : {num2_parity}")
print(f"Proportional Metric : {num1:.2f} is {percentage:.2f}% of {num2:.2f}")
print(f"="*40)

```

Execution Terminal Output

```

=====
ARITHMETIC CALCULATIONS
=====
Addition : 25.00
Subtraction : 11.00
Multiplication : 126.00
Division : 2.5714
Floor Division : 2.00
Modulus : 4.00
Power Elevation : 34012224.00
-----
First Number Parity : Odd
Second Number Parity : Odd
Proportional Metric : 18.00 is 257.14% of 7.00
=====

```

Task 3: Data Type Explorer

Demonstrates data structural casting frameworks across explicit memory classifications while applying dynamic slice indices onto long-string scalar references.

Execution Terminal Output

```

v_int value: 42, type: <class 'int'>
v_float value: 3.1415, type: <class 'float'>
v_str value: 'Python core framework programming module solutions', type: <class 'str'>
v_bool value: True, type: <class 'bool'>
v_none value: None, type: <class 'NoneType'>
Int to Float: 42.0, Type: <class 'float'>
Int to Str: '42', Type: <class 'str'>

```

```
Float to Int: 3, Type: <class 'int'>
Float to Str: '3.1415', Type: <class 'str'>
Method upper() : PYTHON CORE FRAMEWORK PROGRAMMING MODULE SOLUTIONS
Method lower() : python core framework programming module solutions
Method title() : Python Core Framework Programming Module Solutions
First Character : P
Last Character : s
Index 3 Char : h
Index 2 to 7 : thon c
```

Task 4: String Processing Pipeline

This section parses un-sanitized layout inputs into clean data variables by implementing composite chain sequences of trimming, parsing, partitioning, and transformation actions.

Execution Terminal Output

```
=====
EXTRACTED DATA SUMMARY
=====
Full Name Variable : Jean Pierre Mugisha
Formatted Name : Mugisha Jean Pierre
Calculated Age : 28 years old
Extracted City : Kigali
Extracted Country : Rwanda
Major Bounding : Computer Science
Contains 'Science' : True
Metric Score GPA : 3.72
=====
```

Task 5: Financial Calculator

A macro financial calculation component running compound value estimations against structural savings variables while outputting formatted metrics containing currency commas.

Execution Terminal Output

```
=====
FINANCIAL SUMMARY ANALYSIS REPORT
=====
Initial Principal Base : 2,000,000.00 RWF
Simple Interest Yield : 480,000.00 RWF
Total Simple Balance : 2,480,000.00 RWF
Compound Interest Yield : 508,800.00 RWF
Total Compound Balance : 2,508,800.00 RWF
Delta Interest Margin : 28,800.00 RWF
Cumulative Matrix Expenses : 1,200,000.00 RWF
Net Period Savings Metric : 1,308,800.00 RWF
=====
```